Administered Menus and Hidden Clearing: The Microstructure of the Market for Machine Intelligence

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Abstract

Open-weight models make it possible for many firms to sell inference under the same model label. Yet the resulting market is not a conventional spot market: providers post sticky token-price menus, a router privately clears each request, and displayed quotes can be refused when selected. We construct a purpose-built panel of provider-model quotes at five-minute frequency, router-published operational aggregates, historical quote records, and randomized one-token routing probes. Two market facts and one power-gated operational design characterize this market. First, prices are administered menus. Only 2.8% of provider-model-days reprice, changes are large, 93% of quotes lie on cent grids, and state-dependent price gaps predict changes out of sample while congestion does not. Second, prices exhibit exact atoms. The two cheapest quotes tie on 45.9% of multi-provider model-days, versus 13.4% in an independent grid-constrained null. A selected-tie author match is non-discriminating: a post-freeze audit finds 90.0% observed versus 91.9% under a conditional random-label benchmark. The harder all-market identity null is also negative. Authors occupy a shared price in 54.3% of 94 markets versus 53.5% for a random endpoint preserving each market's exact price multiset ($p = 0.466$); the author excess is 0.79 points with author-cluster interval [-9.0,16.1]. Among 196 isolated revisions, 20.4% land exactly on a rival's strictly prior quote, but a scale- and menu-frequency-matched common-menu null predicts 13.4%; the 7.0-point excess has model-cluster interval [-23.1,12.2]. An independently timestamped same-provider control removes own cross-model menu reuse: the residual rises to 13.4 points but its cluster interval still crosses zero. In 6,250 preregistered simulations, the conditional-design test has 3.5% one-sided size but only 43% interpolated power at the observed effect; under a known-clock asynchronous menu it is conservative enough to have zero power even at 50% reactive replacement. All four empirical calibration diagnostics lie outside that mechanism's registered 5th–95th percentile ranges, so it is a stress-test counterexample rather than a fitted model of the market. Third, a randomized-order crossover provides interim evidence that the router manufactures firmness from revocable quotes. In 76 complete four-policy blocks, all-position default routing succeeds on 75 requests while three pinned single-provider policies succeed on 61–64; 76 additional default-only monitoring requests on different models are excluded. These matched block rates are secondary because earlier probes can affect later probes. Carryover-robust first-position outcomes remain masked until every arm reaches 500. A steering audit provides the mechanism link: conditional on currently being cheapest, providers that cut price in the prior week receive 3.9% default selection share versus 23.3% otherwise. Thus the observed rule does not reward past undercutting in the manner proposed to discipline pricing algorithms. We also establish a nonparametric observational-equivalence result: neither within-bin sequencing nor exact landing on a strictly prior quote distinguishes scheduled asynchronous menu revision from rival response without restrictions on the common menu. The resulting sharp identified sets are $[0, 68.6\%]$ for events caused by a unique named prior rival and $[0, 20.4\%]$ for exact strategic landings. The combined evidence supports a dealer-and- dispatcher analogy rather than an automated-market-maker analogy: money prices are sticky, quantity and eligibility clear at request time, and the platform's substitution and prominence rules are first-order market-design instruments.

1 Introduction

The supply of machine intelligence has separated from model ownership. A model author releases weights or an API-compatible specification, independent inference providers host the same model, and a routing platform presents those providers as substitutable sources. A user or application chooses the model label; the router then chooses a provider, retries, or falls back. This vertical separation creates a market whose economic object is not simply "which model is best?" but "how does a platform clear multiple suppliers of the same model?"

That question is difficult to answer from posted prices alone. The price surface is public, but the allocation rule and capacity state are private. The same cheap quote can be visible to every user while being unavailable to a particular request. Conversely, a router can deliver a high fill rate even if no individual quote is firm, by substituting across providers. Treating the public surface as an order book therefore confuses displayed liquidity with delivered liquidity. Treating the router as a decentralized-exchange aggregator makes a related mistake: an inference request is not atomically split across providers, the price is not an executable commitment, and selection depends on hidden eligibility and quality.

We characterize the market with a measurement design that joins four kinds of evidence: high-frequency quote surfaces, router-side operational aggregates, historical price records, and our own randomized micro-purchases. The design keeps three estimands separate. Public observations measure the displayed menu; published operational aggregates measure congestion and enforcement; and owned probes measure the realized policy for one account and workload. None alone identifies market-wide flow, private capacity, or provider cost.

The paper's contribution is two promoted price-surface facts, a falsification ladder that separates exact atoms from author identity and strategic following, a preregistered firmness design with an interim block screen, one mechanism audit, and two observational-equivalence results.

1. **Administered menus.** Prices move infrequently and by large increments. A nested discrete-time hazard rejects a constant-hazard benchmark [Calvo, 1983]: duration and calendar effects matter, and the provider's price gap predicts repricing both in sample and on later days. Rival changes improve the in-sample fit but not yet out-of-sample discrimination; congestion variables do not improve the hazard. The money price is therefore a sticky menu, not the fast-clearing state variable.
2. **Exact price atoms, not identified anchors.** Minimum-price ties are too common to be a rounding artifact. An independent grid-resampling null reproduces the observed price grid and heterogeneity but yields only a 13.4% tie rate, compared with 45.9% in the data. Conditioning on a tie makes the old 90.3% author-match statistic mechanical. An adjacent-grid placebo also fails: although it gives a 53.3-point atom at the author price, an endpoint-label null preserving the exact realized price multiset gives only a 0.79-point author excess and cannot reject exchangeability. Likewise, isolated revisions land on lagged rival quotes much more often than on adjacent grid points, but not significantly more often than on a tightly matched common public price menu. These paired failures are the empirical counterpart of our asynchronous-menu equivalence theorem and prevent an atom from being promoted as author salience, strategic following, or collusion. A preregistered negative control and two size–power experiments then identify the opposite inferential danger: the hard null controls false positives but can have a strictly positive response-detection threshold. We characterize that threshold analytically and distinguish nonidentification from evidence of no response.
3. **Power-gated manufactured firmness.** The money quote barely moves while operational state changes frequently. A prospective crossover randomizes the order of default routing and three pinned-provider policies within model-time blocks. In the 76 complete blocks, all-position

default success is 98.7% while pinned success is 80.3–84.2%. The common rejection level survives randomization of order, but an earlier apparent price-rank gradient does not. These are secondary matched-block rates: the primary first-position outcomes remain masked until 500 assignments per arm. The design therefore suggests, but does not yet promote, router-manufactured firmness.

4. **Steering memory.** Johnson et al. [2023] show that prominence rules which reward past price cuts can disrupt algorithmic collusion. Our revealed-policy audit asks whether current cheapest providers receive more selection after a recent cut. They receive less. This is not proof of collusion–eligibility remains partly hidden–but it rejects the sign of the proposed pro-competitive implementation in the observed sample.

These facts jointly imply a sharper market analogy. The model is a product standard or protocol; an inference provider is a dealer with a revocable quote and hidden inventory; the router is a dispatcher and delegated purchasing agent; and the harness is an application-side broker controlling retries and fallback. The closest mature markets are request-for-quote dealer markets and platform procurement, not constant-function automated market makers. The analogy matters because market quality depends on admission, substitution, and steering rules, not only on quoted spreads.

Our results contribute to three literatures. Work on language-model routing optimizes quality-cost choice across models Chen et al. [2023], Ong et al. [2024], Hu et al. [2024]; we instead measure competition and execution among providers of the same model. Brown and MacKay [2023] show how heterogeneous pricing technology changes competition; we provide the first high-frequency pricing-technology screen for inference providers, while making its short-panel limits explicit. Platform design work studies prominence and algorithmic pricing [Johnson et al., 2023]; our audit turns its comparative static into a statistic that a key-holder can reproduce from realized selections.

2 Institutional setting and economic objects

2.1 The transaction

A request names a model m, supplies an input, and may specify provider preferences. For provider i at time t, the displayed menu contains an input token price p^{in}_{imt} and an output token price p^{out}_{imt}. For a fixed workload with a input and b output tokens, the displayed money quote is

$$q_{imt}(a, b) = a p^{\text{in}}_{imt} + b p^{\text{out}}_{imt}. \tag{1}$$

The platform can additionally observe health, throughput, latency, context-window compatibility, account-specific limits, and contractual eligibility. Let $e_{imtu} \in \{0, 1\}$ denote eligibility for user u, and let c_{imt} denote unobserved deliverable capacity. The default router selects a provider according to an unknown score

$$i^*_{mtu} = \arg\max_i S(q_{imt}, e_{imtu}, c_{imt}, z_{imt}, h_u), \tag{2}$$

where z collects measured quality and h_u is account state. A pinned request sets the candidate set to one provider and can disable fallback. Comparing default and pinned execution therefore measures the value of delegation, not merely a price effect.

The request is served by one provider. Fallback may create multiple attempts, but this is sequential substitution rather than splitting one prompt across a portfolio. For the interfaces studied here, the route cannot generally be observed as a firm executable allocation before purchase. This absence of a public pre-trade route is why literal order-flow front-running is not identified from public data.

2.2 Measurement levels

We use an explicit observability hierarchy. It prevents a common error in this market: assigning the identifying content of realized flow to a price-only simulation.

Level	Object	Identified use
L0	Public quote surface	Price levels, gaps, ties, grids, and repricing events
L1	Documented policy simulation	Mechanical ranking or allocation proxy only
L2	Public operational aggregates	Congestion, rate-limit, derank, and enforcement state
L3	Owned realized requests	Selected provider, success, latency, cost, and fallback for the probed account
L4	Provider or router logs	Capacity commitments, full attempt order, and cross-user flow

Table 1: Observability hierarchy. The core paper uses L0–L3. No claim that requires cross-user order flow or private provider cost is made.

Public token-allocation displays sometimes expose a deterministic price-weighting proxy. When a documented inverse-square proxy is available, we compute

$$\widehat{s}_{imt} = \frac{q_{imt}^{-2}}{\sum_j q_{jmt}^{-2}}, \qquad \frac{\partial \log \widehat{s}_{imt}}{\partial \log q_{imt}} = -2(1 - \widehat{s}_{imt}). \tag{3}$$

Equation (3) is a policy arithmetic identity. It is not demand elasticity and is never substituted for realized routing in our causal claims.

3 Data and research design

3.1 Capture system

The live panel snapshots public provider-model menus every five minutes. Each record stores a UTC capture time, stable provider and model identifiers, input and output token prices, capacity ceilings when displayed, and router-published five- and thirty-minute enforcement aggregates. Captures are append-only and mirrored to remote object storage. The analysis also uses a three-year historical quote registry for lifecycle checks, public model-level token-allocation records, and owned generation records for requests made with our study key.

The pricing freeze used in the main tables contains 2,968,449 endpoint snapshots from 1,805 capture runs over nine days. It covers approximately 300 models and 69 providers with enough exposure for provider-level cadence classification. The all-field event ledger contains 3,212 changes; the frozen Brown–MacKay screen retains 274 positive completion-price events over nine observed quote days. The public routing simulation contains 562,550 provider-model-time rows over six days. Owned legacy traffic contains 2,659 raw rows over 65 runs; after deduplication and exclusion of outcome-masked studies, 716 descriptive attempts remain and 617 identify the selected provider. The randomized crossover is a separate prospective release.

Panel	Rows / attempts	Span	Primary role
Endpoint snapshots	2,968,449	9 days	Prices, grids, gaps, enforcement state
All-field price events	3,212	9 days	Repricing incidence and event construction
Completion-price events	274	9 quote days	Pricing-technology cadence screen
Public allocation proxy	562,550	6 days	Mechanical share-price comparative statics
Legacy owned attempts	716	65 runs	Descriptive provider-selection audit
Randomized crossover	304 + 76	prospective	76 matched four-policy blocks plus default-only monitoring
Historical quote registry	multi-year	3 years	Lifecycle and duration sensitivity

Table 2: Frozen analysis panels. Sample sizes refer to the corrected nine-day manuscript release and are updated only at registered promotion gates.

3.2 Cleaning and event construction

Prices are converted to dollars per million tokens before comparisons. We use standard provider variants and remove exact duplicate captures. A price event is the first observation at which either token price differs from the preceding observation for the same provider-model pair. Five-minute timing creates interval censoring within a capture bin; we do not infer within-bin order. The first price spell in a panel is left-censored and is excluded from moments that require age.

Operational counters are router-published aggregates, not raw request logs. Their reporting can be intermittent. Regressions that use utilization or rate-limit state include a missingness indicator and are repeated on the complete-reporting subsample. Status-derived incidents are labeled as heuristics unless the source publishes exact incident timestamps. Owned probes record only metadata needed for the study–assignment, policy, provider, latency, token count, cost, and status–and never retain prompt payloads.

3.3 Claim discipline

The empirical sequence was adaptive, so we distinguish discovery from confirmation. Early descriptive analyses generated the administered-menu, price-atom, and firmness hypotheses. The fixed-order firmness pilot revealed a large default-pinned gap but confounded policy with position. We disclosed that failure, froze a randomized-crossover replacement before observing its outcomes, and treat the first release as interim. Dynamic pricing results are re-estimated at a registered 30-day gate using both the original and expanded vintages. The endpoint-label and matched-menu nulls were developed after inspecting the frozen panel, are labeled exploratory there, and are now fixed for that 30-day release. Tests that have not reached their registered support are reported as power-gated rather than as null findings.

4 Fact 1: prices are administered menus

4.1 Frequency, size, and the clock

Only 2.8% of provider-model-days contain a price change. Across completed endpoint spells, the implied average duration is 26.7 days. Conditional on a change, the median absolute log change is 0.134; two thirds are cuts. Ninety-three percent of quotes lie on cent-per-million-token grids. Twentysix percent of cuts occur at 00:00 UTC, a concentration difficult to reconcile with a continuously clearing spot price and consistent with scheduled menu updates.

Raw pooled price-change kurtosis is 113 because providers and model scales differ. After standardizing changes within provider, kurtosis is 3.54; the provider median is 3.4 with interquartile range 2.1–4.9. This statistic lies between a simple state-dependent menu-cost benchmark and the Calvo benchmark discussed by Alvarez et al. [2016]. We use it as a sufficient-statistic diagnostic, not as a structural estimate: the short panel leaves many providers with too few changes for stable within-unit moments, and observation costs can mimic infrequent adjustment Alvarez et al. [2011].

4.2 A nested repricing hazard

For provider-model pair k on day d, let $Y_{kd} = 1$ if a price changes. We fit a nested logit ladder

$$\mathbb{P}(Y_{kd} = 1 \mid X_{kd}) = \Lambda\big(\alpha + T_{kd} + G_{kd} + C_{kd} + R_{kd}\big), \tag{4}$$

where T contains age and calendar terms; G contains the absolute and signed gap from the cheapest rival plus GPU-cost changes; C contains published utilization and rate-limit state; and R contains rival raises and cuts in the prior day. The ladder adds these blocks sequentially. Provider activity is included as a frailty proxy, but with only nine days we do not estimate a full unit-frailty duration model.

The constant-hazard model is decisively rejected by duration and calendar terms. Adding price gaps improves fit ($p = 1.0 \times 10^{-5}$); adding congestion does not ($p = 0.17$); adding rival moves improves in-sample likelihood ($p = 0.0018$). Out-of-sample results are more disciplined. On a day split, time dependence alone has AUC 0.525 and the state-dependent model has AUC 0.638. The strategic model has AUC 0.555 on 36 test events. Thus the gap channel survives temporal validation; the rival-reaction channel does not yet generalize.

Hazard rung	Added state	Incremental evidence	Day-split AUC
Constant	none	baseline	0.500
Time-dependent	duration, clock	rejects constant hazard	0.525
State-dependent	own-rival gap, GPU change	LR $p = 1.0 \times 10^{-5}$	0.638
Congestion	utilization, rate limits	LR $p = 0.17$	not promoted
Strategic	rival moves	LR $p = 0.0018$ in sample	0.555

Table 3: Nested repricing hazard. The strategic rung is labeled in-sample-only; the 30-day both-vintage gate is preregistered.

The appropriate conclusion is narrower than "providers use pricing algorithms." Prices respond to their distance from the menu's competitive reference point and are updated on scheduled attention dates. We call them administered menus because the money quote is set periodically while the physical state clears elsewhere. Whether rival reactions reflect common costs, algorithmic competition, or coordinated behavior remains unresolved.

5 Fact 2: minimum prices exhibit exact atoms

5.1 The tie atom and its null

For each model-day with at least two providers, order the completion quotes and define the relative second gap

$$g_{md} = \frac{q_{(2),md} - q_{(1),md}}{q_{(1),md}}. \tag{5}$$

Across 1,311 model-days, $g_{md} = 0$ in 45.9%. A zero gap is especially common in two-provider markets, but the atom persists in thicker markets. Among all non-minimum endpoint observations, 47.4% are exactly tied at the minimum; the remaining distribution has relatively little mass in very small undercuts.

Rounding can generate ties mechanically. We therefore construct an independent grid null. Within each model-day, each provider draws a log-price deviation from the pooled empirical distribution of deviations from model-day medians. Draws are independent across providers, added to the observed model-day median, and snapped to the observed cent-per-million-token grid. We preserve the observed number of providers by model-day and repeat the procedure. The null produces a 13.4% mean tie rate with 1.0 percentage-point simulation standard deviation. Snapping to a coarser dime grid increases the null to 27.6%, still well below 45.9%. Because the pooled deviation distribution retains some empirical tie mass, this null is conservative with respect to independent coordination on the same grid point.

Post-freeze robustness (1,456 model-days, 148 model clusters) retains a 45.1% tie rate. The cent-grid null is 12.9%, giving a 32.2-point excess (model-cluster bootstrap 95% interval [25.8, 37.8]); the dime-grid null is 27.2%, giving an 18.0-point excess [9.7, 25.1]. The bootstrap resamples whole models and re-simulates the null within each draw; it is a robustness refresh, not a replacement for the frozen v3 vintage.

Statistic	Estimate	Denominator
Observed tie at minimum	45.9%	1,311 model-days
Independent cent-grid null	13.4%	matched simulations
Independent dime-grid sensitivity	27.6%	matched simulations
Observed / cent-null ratio	3.4	–
Observed / dime-null ratio	1.7	–

Table 4: Minimum-price tie atom. The null preserves market thickness, observed deviation heterogeneity, and a discrete price grid, but breaks cross-provider dependence.

5.2 Identifying the focal level

The selected-tie identity statistic needs its own null. If a model has n providers, t are tied at the minimum, and r author-operated labels are placed exchangeably among provider slots, then the conditional probability that an author label lies in the tied set is

$$1 - \frac{\binom{n-t}{r}}{\binom{n}{r}}. \tag{6}$$

It equals one when all providers tie. Thus conditioning on a tied model can make an author match nearly automatic. In the post-freeze audit of the frozen panel's final snapshot, the robust author-family crosswalk produces 45 matches among 50 selected tied models, or 90.0%. Summing equation (6) across those markets predicts 45.97 matches, or 91.9%; the exact Poisson-binomial upper-tail probability is 0.913. The previously emphasized 65-of-72 frozen statistic is therefore demoted: its high level does not identify the author's price as focal.

The all-market estimand avoids selecting on a tie. It starts from every multi-provider model with one author-operated and at least one third-party endpoint. At the primary dime-per-million-token spacing, 51 of 94 models, or 54.3%, have a third-party quote exactly at the author price, versus 1.5% at 40 adjacent placebo levels. The resulting 52.8-point contrast is precisely estimated but answers only whether an observed price level contains mass. Empty nearby grid points are not an identity null when providers use a common discrete menu.

The harder null preserves each model's entire realized price multiset and places the author label uniformly among its endpoint slots. An endpoint is "shared" when at least one other endpoint has the same price. If s_m of n_m endpoints are shared, the random-anchor probability is s_m/n_m. Authors occupy a shared price in 54.3% of models, while the summed random-anchor expectation is 50.26 of 94, or 53.5%. The exact Poisson-binomial upper-tail probability is 0.466. The mean author-minus-random-anchor difference is 0.79 points with a 20,000-draw author-cluster interval of [-9.0,16.1] points and leave-one-author-out range [-1.9,6.5]. Comparing author–third-party equality with equality among third parties gives -0.94 points over 48 eligible models, with interval [-8.8,12.0]. Thus the atom is real, but author identity is not special under a price-multiplicity-preserving null.

Identity statistic	Estimate	Benchmark
Selected-tie author match	90.0%	91.9% random label
All-market author at shared price	54.3%	53.5% random endpoint
Author-minus-random endpoint	0.79 pp	[-9.0,16.1] cluster interval
Author-vs-third-party pair density	-0.94 pp	[-8.8,12.0] cluster interval

Table 5: Price-multiplicity-preserving identity audit. Both conditional and all-market author statistics fail their exact endpoint-label nulls. The post-freeze specification is fixed for the 30-date release.

The same problem arises in time series. We isolate consecutive snapshots no more than 15 minutes apart in which exactly one provider changes price and all compared quotes are observed on both sides. Of 196 revisions across 18 models and 15 providers, 40, or 20.4%, land exactly on a rival's strictly prior quote. Only 1.0% land on the average of 40 adjacent placebo levels, producing a seemingly large 19.5-point excess. But common price menus can make the adjacent levels empty by construction.

For each event, our matched-menu null draws the same number of prices as observed rivals from other models in the immediately prior snapshot, restricted to prices within a factor 1.25 of the mover's new quote. The exact hypergeometric hit probability preserves local price scale, risk-set size, and the global frequency of each exact menu point. It predicts 13.4% matches. The remaining 7.0-point excess has model-cluster interval [-23.1,12.2] and provider-cluster interval [-22.4,18.2]; its leave-one-model-out range is [-10.4,9.7]. One model supplies 73.5% of events and 85.0% of exact landings. Wider control bands mechanically lower the menu baseline, so the registered replication retains the most local 1.25-band null. The current panel does not distinguish strategic following from a common asynchronously refreshed menu. A past-only same-model sensitivity with a 24-hour washout is more favorable: 189 events have a 6.8% historical-menu baseline and 14.4-point excess. Yet its model-cluster interval is [-0.1,17.5] points, while its positive provider-cluster interval is driven by the same concentrated models and movers. We register this SKU-specific comparison as secondary rather than selecting it over the tougher primary null.

The preregistered same-provider/across-model control was committed before its estimate was computed. Among 175 comparable events, only 12, or 6.9%, reuse the mover's own price on another model at the strictly prior timestamp. Own-menu support is 5.3% among exact rival landings and 7.3% otherwise, a −2.0-point association with model-cluster interval [-31.0,0.9]. Removing those 12 events leaves 163 own-menu-novel revisions: their exact landing rate is 22.1% against an 8.6% matched-menu benchmark. The 13.4-point residual is larger, and every leave-one-model-out estimate is positive [1.4,15.2], but its model-cluster interval [-11.5,18.5] still fails the frozen promotion rule. The negative control therefore weakens a simple provider-template explanation without identifying rival response.

Lagged-landing statistic	Estimate	Benchmark or interval
Isolated continuous revisions	196	18 models, 15 providers
Exact landing on lagged rival quote	20.4%	40 events
Adjacent-grid placebo	1.0%	19.5 pp excess
Matched common-menu probability	13.4%	factor-1.25 risk set
Exact-minus-common-menu	7.0 pp	[-23.1,12.2] model-cluster interval
Past-only same-model menu	14.4 pp	[-0.1,17.5] model-cluster interval
Own-provider cross-model support	6.9%	−2.0 pp landing association
Own-menu-novel common-menu excess	13.4 pp	[-11.5,18.5] model-cluster interval

Table 6: A weak placebo suggests price following; the matched common-menu null does not. The table is an exploratory frozen-panel audit registered unchanged for the earliest 30-date release.

Proposition 1 (Asynchronous-menu observational equivalence)**.** *Let a finite snapshot panel take values in a discrete public price menu. Suppose one-provider revisions can be scheduled by provider-specific refresh clocks and selection rules driven by a latent public menu state, without dependence on rival quotes. A model that instead permits rival-triggered revision is also admissible. Then any observed exact cross-sectional match or exact landing on a strictly prior rival quote can be generated by both models. If L of N isolated revisions land exactly, the share of all revisions that are exact strategic landings is sharply identified only within* $[0, L/N]$.

The constructive proof is in section B. For the frozen panel, $L/N = 40/196 = 20.4\%$. The matched-menu test restricts the otherwise unrestricted null by imposing exchangeability within a local public risk set; its failure to reject is empirical, while proposition 1 explains why the weak adjacent-grid result cannot carry a behavioral interpretation.

The hard null can also fail by being too conservative. Let p_e denote an event's nonreactive exact-landing probability and q_e its matched-menu benchmark. Suppose a response, when it occurs, forces an exact rival-price landing.

Proposition 2 (Detection threshold of a dominating menu null)**.** *If reactive replacement occurs with probability ρ, independently conditional on the event risk set, then*

$$\mathbb{E}[Y_e - q_e] = p_e - q_e + \rho(1 - p_e). \tag{7}$$

When $q_e \geq p_e$, the matched-menu residual has correct one-sided sign under the nonreactive mechanism but becomes positive only if

$$\rho > \rho_e^\star \equiv \frac{q_e - p_e}{1 - p_e}. \tag{8}$$

For a common ρ over a fixed event panel, replace (p_e, q_e) by their event means. Thus conservativeness and power loss are two implications of the same benchmark dominance.

The frozen-design simulation holds each event's q_e and model cluster fixed. In 1,000 replications per response level, one-sided false promotion is 3.5%; power is 29.7% at $\rho = 0.05$, 51.9% at 0.10, and 94.7% at 0.25. The observed 7.0-point excess corresponds to $\rho = 0.081$ in that conditional mixture and only 43.4% interpolated power. A second experiment generates 250 complete 576-tick panels per level from public menus and known provider refresh clocks. At $\rho = 0$ the exact-landing rate is 59.9% while the global-menu benchmark is 90.4%; the implied fixed-panel threshold is approximately 0.762. The unchanged test never promotes, even at $\rho = 0.50$. This is not evidence that the market has a large or small ρ; it shows that failure to reject the hard null cannot be read as evidence of no strategic behavior.

Two preregistered robustness disclosures sharpen that boundary. Exact enumeration of all 2^{18} model-cluster sign assignments gives one-sided $p = 0.425$ for the full residual; enumeration over the 14 own-menu-novel clusters gives $p = 0.214$. The positive event-weighted estimates therefore lack broad cluster support. The known-clock family is also not empirically calibrated. Its null 5th–95th percentile ranges are 1,228–3,381 events, 49.0–73.2% exact landings, 89.1–91.3% menu probabilities, and −41.5 to −17.3 points of residual. The empirical values—196, 20.4%, 13.4%, and +7.0 points—fall outside all four ranges. Accordingly SIM2 is a stress-test construction showing possible benchmark dominance, not a fitted data-generating process or an estimate of ρ.

Calibration statistic	Empirical	SIM2 p05	SIM2 median	SIM2 p95
Extracted events	196	1,228	2,150	3,381
Exact landing share	20.4%	49.0%	58.8%	73.2%
Matched-menu probability	13.4%	89.1%	90.5%	91.3%
Exact-minus-menu residual	+7.0 pp	−41.5 pp	−31.5 pp	−17.3 pp

Table 7: Frozen empirical statistics against the registered $\rho = 0$ SIM2 distribution. Every empirical value is outside the simulated 5th–95th percentile range; no simulation parameter was retuned.

Direct provider APIs and router quotes are at exact parity in 99.6% of matched cases; we see almost no systematic direct-channel undercut. Of 31 isolated tie formations, 74.2% are formed by a downward move, and 88.4% of 43 isolated tie breaks are downward. Those directions are competitive. The evidence supports shared discrete price menus, not a named anchor: common cost conventions, provider templates, strategic matching, and a Schelling-like device remain observationally compatible.

6 Interim design: does the router manufacture quote firmness?

6.1 Price state and operational state

Over the pricing panel, only 12% of endpoints ever change price while 86% experience variation in published rate-limit state. At thirty-minute horizons, latency and enforcement variables load much more strongly on operational changes than price does. Raises tend to occur after slack rather than at congestion spikes. These facts suggest that scarcity is cleared through quantity, eligibility, and fallback while the menu remains fixed.

Displayed unavailability is not enough to establish quote revocability: public counters need not describe the study account. We therefore purchase minimal one-token completions under four policies. Default routing delegates provider choice and permits fallback. Pinned-cheapest, pinned-second, and pinned-random restrict the request to one provider selected from the contemporaneous public surface.

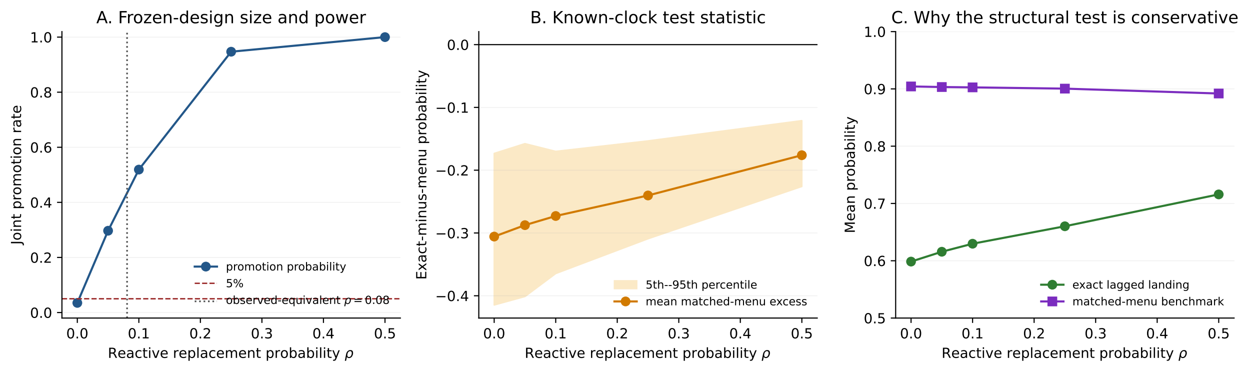

Figure 1: Preregistered size and power. Panel A conditions on the frozen 196-event design. Panels B–C run the unchanged statistic on known-clock asynchronous-menu panels with nested reactive replacement. The global-menu benchmark controls false positives but can dominate the focal rival set enough to eliminate power. The registered mechanism fails the empirical calibration audit and is interpreted as a stress test.

6.2 The failed pilot and randomized replacement

The first 96 complete blocks always ran default first. It found a 17.7 percentage- point default advantage over the cheapest pinned quote, but order and policy were perfectly confounded. Account-level throttling, warm-up, or within-block interference could produce the entire gradient. We retain that result only as a design failure.

The replacement randomizes both model order and policy order for every block and records the assignment seed before execution. Let Z_{bp} be assignment of policy p to first position in block b, and Y_{bp} request success. Under random assignment, the first-position difference

$$\widehat{\tau}_p^{(1)} = \frac{\sum_b Z_{bD} Y_{bD}}{\sum_b Z_{bD}} - \frac{\sum_b Z_{bp} Y_{bp}}{\sum_b Z_{bp}} \tag{9}$$

is design-unbiased despite arbitrary effects of the first probe on later probes. Pooled within-block contrasts are more precise but require that carryover is controlled by randomized order and is not policy-specific after conditioning on position. We therefore report both and make the first-position estimator the carryover-robust estimand at the promotion gate.

The frozen interim screen contains 76 complete four-policy blocks. Every policy therefore has 76 matched, randomized-order attempts: default succeeds on 75 (98.7%), pinned-cheapest on 61 (80.3%), pinned-second on 62 (81.6%), and pinned-random on 64 (84.2%). The collector also made 76 default-only monitoring requests on four additional models. Pooling those rows would produce 151/152, but it would confound policy with model support; we exclude them. Marginal Wilson intervals in figure 2 are descriptive. Random order balances clock position, but treatment-specific carryover can still affect later positions.

The carryover-robust first-position analysis is outcome-masked until every arm reaches 500 assignment-verified observations. At the 2026-07-16 20:00 UTC outcome-free audit, the default, cheapest, second, and random counts are 33, 37, 34, and 36, with 100% seed-replay success. The eventual release uses the earliest chronological prefix satisfying the gate, Holm-adjusted default-pinned randomization tests, and a prespecified rank-gradient test. Until then, the defensible interim conclusion is narrower: the approximately 19% all-position pinned rejection level survives randomization of order and the cheapest/second/random gradient is flat, but router-manufactured firmness is not yet a promoted causal fact.

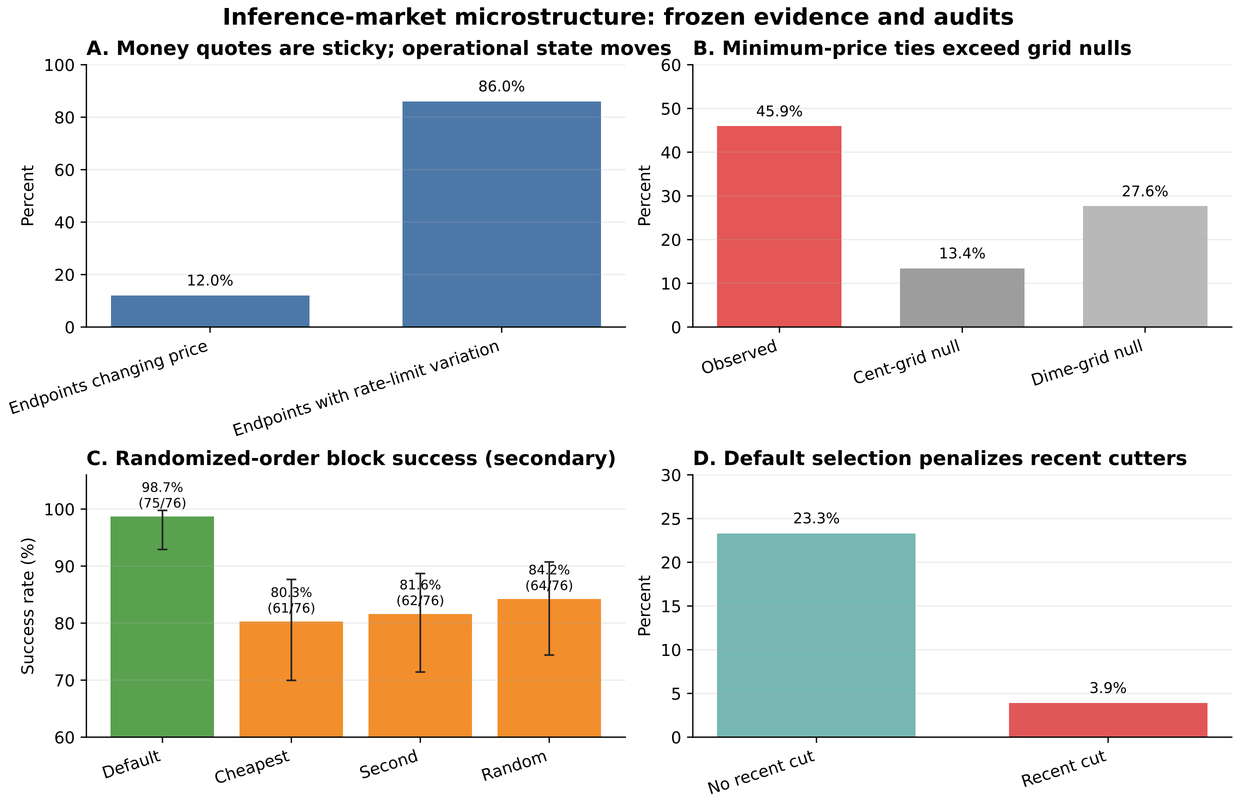

Figure 2: Frozen evidence and audits. Panel C shows secondary all-position rates from 76 complete randomized-order blocks with marginal Wilson 95% intervals; the carryover-robust first-position outcomes remain masked until 500 assignments per arm. Panels A, B, and D are descriptive panel statistics.

7 The steering audit

The platform-design problem is not exhausted by current-price ranking. Johnson et al. [2023] study dynamic price-directed prominence: a platform can reward a seller that previously cut price with additional exposure, changing the repeated-pricing game even after the current ranking is held fixed. We turn the sign of that comparative static into a revealed-policy audit.

For provider-model-day cell (i, m, d), let S_{imd} be its share of selected providers among our default probes, C_{imd} indicate a price cut in the prior seven days, and R_{imd} denote its current price rank. The audit estimates

$$S_{imd} = \alpha_{md} + \beta C_{imd} + f(R_{imd}) + \varepsilon_{imd} \tag{10}$$

and reports the nonparametric cheapest-provider cells. Among 251 provider-model-days, a currently cheapest provider without a recent cut has mean selection share 23.3%; one with a recent cut has 3.9%. The rank-controlled coefficient is negative. The observed policy therefore does not implement the positive past-cut reward that disciplines patient pricing algorithms in the theory.

Eligibility is the central threat. Publicly quoted endpoints may be ineligible for our region, prompt, or account. Restricting the comparison to provider-model pairs whose eligibility is independently confirmed by pinned probes preserves the direction–30.0% without a recent cut versus 9.5% with a cut–but leaves only 11 and one cells. We therefore label the unrestricted result an audit statistic, not a causal effect of cutting price on future prominence. Reversal would require eligibility misclassification to be concentrated among recent cutters, a pattern the prospective pinned panel is designed to measure.

Two interpretations remain. Recent cuts may predict degraded capacity, causing the router rationally to protect users. Alternatively, a negative-memory rule taxes price competition. A randomized visibility test or router-side eligibility log distinguishes them. What the current data rule out is narrower but useful: for this account and sample, the revealed selection pattern does not reward past undercutting.

8 Bounded secondary evidence

This section reports results that sharpen the market description but do not carry the paper's novelty claim.

8.1 Pricing cadence and Brown–MacKay

Of 69 exposed providers, 19 reprice at least once: nine are classified intraday, seven daily, and three weekly; 50 are inactive or left-censored. Cadence rates divide changes by each provider's observed quote exposure within the nine-day window, rather than by the shorter span between its first and last event. Conditional on model-day, slow providers quote 10.1% more than fast providers. The log-price coefficient is -0.0964 with provider-clustered standard error 0.0281 and 95% interval $[-0.1515, -0.0413]$ across 5,501 observations and 127 models. On the smaller public-quality overlap, the coefficient is -0.3892 with interval $[-0.6374, -0.1411]$.

The association resembles Brown and MacKay [2023], but the proposed reaction mechanism does not yet discriminate from state-dependent pricing. Capture times interval-censor updates, so we exclude simultaneous multi-provider batches as initiators and never order changes first observed in the same snapshot. Freezing cadence on the first 70% of events then leaves 19 unambiguous evaluation waves but zero slow-initiator risk pairs. Among 188 uniquely linked reactions, the temporal holdout RMSE is 0.23835 for the state-only rule and 0.23760 for the Brown–MacKay rule. The paired MSE improvement is 0.000355 with seven-model-cluster bootstrap interval $[-0.000972, 0.008033]$. An exact cluster sign-flip sensitivity gives $p = 0.023$, but the few-cluster interval crosses zero. We therefore call the two rules predictively indistinguishable: the cadence premium is a resolved association, not evidence that slow firms initiate and fast firms respond.

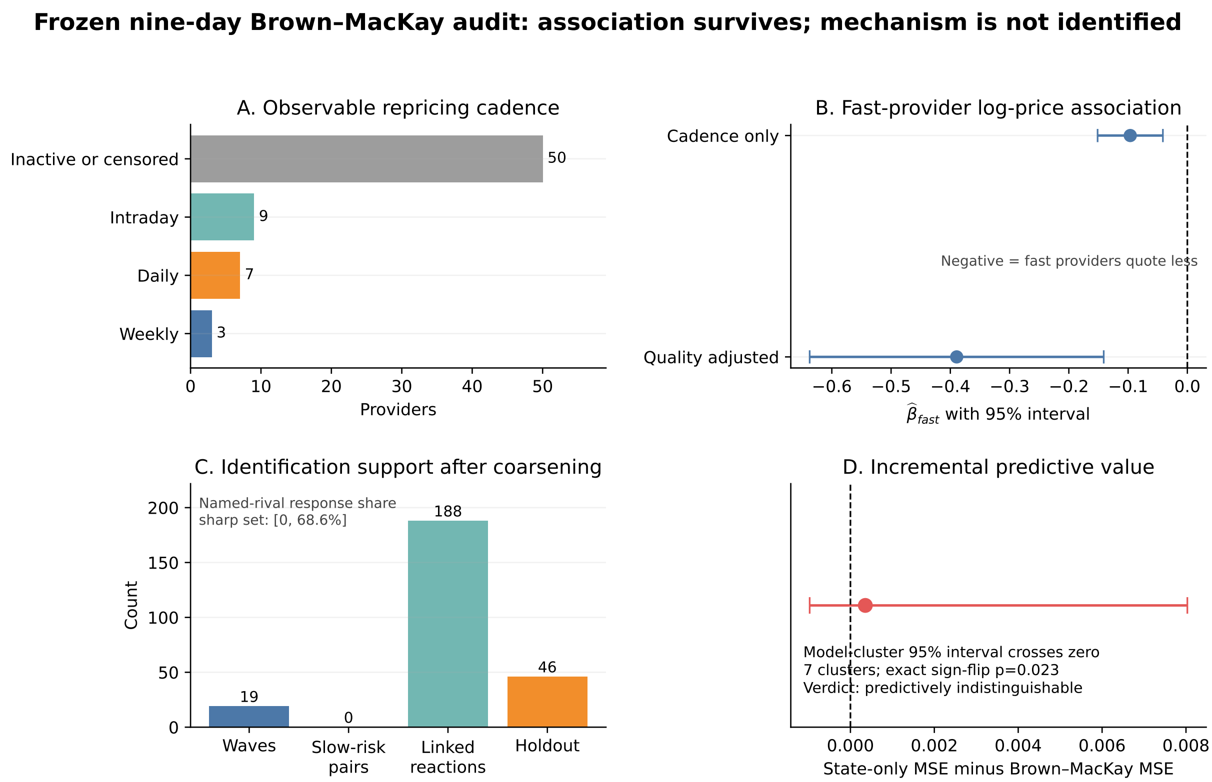

Figure 3: Frozen Brown–MacKay audit. Negative cadence coefficients survive public-quality adjustment, while the slow-initiator support and incremental reaction-rule evidence do not identify the mechanism. The 188 uniquely ordered candidates among 274 events imply a sharp named-rival response-share set of $[0, 68.6\%]$.

The negative reaction result is not merely a small-sample qualification. The observation clock imposes a nonparametric identification limit even with an arbitrarily long panel.

Proposition 3 (Coarsened-timing observational equivalence)**.** *Let latent provider quotes be right-continuous paths observed only at grid times $t_k = k\Delta$. Consider an interval in which at least two providers change between t_{k-1} and t_k. If an administered-menu model permits provider-specific scheduled updates driven by latent public state, and a reactive model permits a provider to update within less than Δ after observing a rival move, then the sampled transition is compatible with both models. Moreover, every changed provider can be made the latent first mover in a reactive path that produces the same sampled quotes. Hence neither within-bin leader identity nor rival-triggered response is point identified from the snapshot panel.*

Corollary 4 (Sharp named-rival response bounds)**.** *For each of N sampled price events, let $U_e = 1$ when the event has a unique most-recent strictly prior rival move inside a declared lookback window, and let $R_e = 1$ when that named move caused the event. The sample response share $\theta = N^{-1}\sum_e U_e R_e$ satisfies*

$$0 \leq \theta \leq \frac{1}{N}\sum_e U_e. \tag{11}$$

Without restrictions beyond proposition 3, both endpoints are attainable.

The proofs are constructive and appear in section B. The result allows unrestricted latent state and response maps, so restrictions on either can restore identification. Without such restrictions, however, more five-minute snapshots improve frequency estimates but do not resolve ordering inside a snapshot. Excluding same-snapshot batches therefore prevents invented order; the uniquely and strictly earlier links used above identify temporal candidates, not causal triggers. Literal quote-surface front-running requires finer update timestamps, an exclusion restriction on common shocks, or randomized visibility. In the frozen nine-day panel, 188 of 274 price events have a unique most-recent strictly prior rival inside 72 hours. Consequently, the share of all price events caused by that named rival is sharply bounded by $[0, 188/274] = [0, 68.6\%]$: candidate abundance narrows the upper endpoint but supplies no positive lower bound.

8.2 Entry and long-memory demand

Across 304 models, the elasticity of quoting-provider count with respect to token demand is 0.149 (95% interval $[0.126, 0.175]$); the active-provider elasticity is 0.173. Both are below i.i.d. square-root and cube-root entry benchmarks. A long-memory correction maps a measured multi-scale count Hurst exponent of 0.835 to $n^{(2-2H)/2} = n^{0.165}$, numerically close to the observed slope. We treat this as a remark, not identification: entry and demand are simultaneous, the panel is short, and a deseasonalized intraday estimator is anti-persistent. The registered within-market interaction is underpowered.

8.3 Retry feedback

Within endpoint, subsequent thirty-minute request growth rises with the current rate-limit share. Demand-controlled OLS gives 0.167, while a backward placebo is -0.126. Instrumenting focal rationing with the same provider's rate limiting on other models gives 0.023 with interval $[-0.043, 0.085]$. The sign-asymmetric OLS is consistent with retry amplification; the instrument bounds the same-endpoint component near zero and excludes the rerouting margin. We do not convert either coefficient into welfare because request aggregates do not reveal user values, backoff policies, or cross-provider substitution.

8.4 Conduct screens

A naive event classifier labels 83% of rival responses to cuts as punish-and-revert. Reconstructing the initiator's subsequent action changes the classification: 42.1% are punish-and-revert candidates and 41.1% are cuts the initiator later withdraws. Among raises, 65.9% are followed within 72 hours, but many initiators have low measured volume. These classifications show why posted- price event shapes alone are not proof of algorithmic collusion. Common cost shocks, experimentation, and sparse observation remain observationally equivalent without an adoption event or exogenous cost shock.

9 Market-design implications and welfare

The facts identify design wedges, not a structural welfare number. Let user type θ have completion value v_θ, delay cost d_θ, and rejection loss ℓ_θ. If provider i serves quantity x_i with capacity μ_i, quality a_i, unit cost $c_i(a_i)$, and delay $W_i(x_i, \mu_i)$, a planner chooses admission A_θ, routing $x_{i\theta}$, capacity, and quality to maximize

$$
\begin{aligned}
\max \quad & \int A_\theta \left[v_\theta(1-r_\theta) - d_\theta W_\theta - \ell_\theta r_\theta\right] \mathrm{d}F(\theta) \\
& - \sum_i \left[k_i(\mu_i) + c_i(a_i)x_i\right] - \int q_\theta(1-a_{i(\theta)})\mathrm{d}F(\theta),
\end{aligned} \tag{12}
$$

subject to capacity, queue stability, and retry feedback. Transfers cancel from global welfare but determine participation and investment.

The decentralized agents solve different problems. A provider chooses a sticky menu, capacity, and hidden quality to maximize margin times routed volume minus capacity and adjustment costs. The router chooses admission, allocation, fallback, monitoring, and steering to maximize fees and retained user value. A harness chooses model, retry, and fallback policies to maximize application retention net of spend. A user sees the harness's delivered price-quality-delay bundle, not the provider market directly.

The evidence isolates four possible failures of decentralization.

1. **Admission.** A high default fill rate can be privately valuable but socially excessive if retries impose congestion on other users. User-value signals are needed to price or defer low-value requests.
2. **Quality verification.** Selection on price and latency can induce unmeasured fidelity shaving. The public panel has no task-level quality certificate, so the corresponding welfare term is unidentified.
3. **Steering.** Prominence is an allocation instrument. The negative past-cut association fails the sign of a known pro-competitive dynamic rule and may reduce the return to undercutting unless it is explained by hidden quality or capacity.
4. **Retry feedback.** The OLS-placebo pattern is consistent with positive feedback, but the IV bound implies little same-endpoint amplification. The important unmeasured margin is rerouting across providers.

The central market-design implication follows from observed quote revocability and the platform's documented substitution role, not from a promoted causal effect. A price-only best-execution standard is incomplete when the quote is revocable. A meaningful score must combine expected money cost with success probability, latency, and fidelity for a declared workload. Capacity certificates, explicit fallback prices, or auditable provider-level fill rates would move the market from displayed menus toward executable commitments. The companion mechanism paper studies such certificates; this paper supplies the empirical reason they are needed.

10 Related work

Model routing. FrugalGPT, RouteLLM, and RouterBench study how to choose among models with different quality and cost Chen et al. [2023], Ong et al. [2024], Hu et al. [2024]. Their unit is a model. Our unit is a provider-model endpoint, holding the model label fixed and measuring the market that supplies it. The two problems compose: a harness may choose the model and a marketplace may then choose the supplier.

Pricing technology and focal points. Calvo [1983] provides the constant-hazard benchmark; observation and menu-cost models motivate duration, gap, and change-size diagnostics [Alvarez et al., 2011, 2016]. Brown and MacKay [2023] show that heterogeneous repricing frequency can soften competition. Knittel and Stango [2003] identify a nonbinding regulatory ceiling as a focal point.

We import their empirical logic and then show why an adjacent-level counterfactual is insufficient in a market with shared discrete menus. Exact endpoint-label and matched common-menu nulls separate the existence of a price atom from claims about anchor identity or strategic following. The latter is a dynamic pricing analogue of the reflection problem: correlated behavior and endogenous response are inseparable without reference-group or shock restrictions [Manski, 1993].

Platform steering and marketplace experiments. Johnson et al. [2023] connect prominence rules to algorithmic pricing. Our steering-memory statistic is an empirical implementation of their sign restriction. Switchback and marketplace experiments require special care because treatment changes congestion and later outcomes Bojinov et al. [2023], Holtz et al. [2025]; our first-position estimator protects the firmness contrast from arbitrary within-block carryover. Online matching under stochastic rewards provides the OR analogue for provider selection Brubach et al. [2025], but the inference setting adds sticky provider menus and hidden eligibility.

Emerging AI-platform evidence. Recent work studies quality-adjusted AI prices and platform self-preferencing Wang [2026], Zhang et al. [2026]. We complement these studies with provider-level microstructure and a reproducible realized-policy audit. The institutional APIs are documented by the platforms themselves OpenRouter [2026], Hugging Face [2026]; all policy claims in this paper are versioned to the captured interfaces rather than generalized to undocumented behavior.

11 Limitations and conclusion

Five limitations define the claim boundary. First, the main pricing panel is nine days, so dynamic hazard and cadence results await the registered 30-day both-vintage re-estimation; the identity and lagged-landing hard nulls are post-freeze and await the same replication, while the local outcome-free calendar contains 10/30 dates. The matched-menu test controls one-sided size in the conditional simulation but has only 43.4% power at the observed effect and can be far more conservative in known-clock panels; its nonrejection is not evidence of zero response. Exact cluster sign-flip p-values are 0.425 and 0.214, and every registered empirical calibration diagnostic rejects the known-clock family as a fitted description. Second, the randomized crossover's all-position block screen is secondary, and its carryover-robust first-position counts are only 33–37 of 500 per arm. Third, owned probes identify one account, region, prompt family, and time window; they do not identify market-wide provider shares. Fourth, public operational fields are router-published aggregates with missingness, not raw attempt logs. Fifth, provider cost, private rebates, task fidelity, and cross-user request order are unobserved. We therefore do not claim literal front-running, collusion, private cost, or a welfare loss.

Within those limits, the evidence changes the economic description of inference markets. Prices are not the rapidly clearing state variable. Providers post sticky administered menus, minimum-price ties exceed independent grid benchmarks, but exact atoms do not survive hard tests of author identity. Strategic following is neither promoted nor ruled out: its hard null is observationally defensible but underpowered at the observed effect. The secondary block screen is consistent with the platform manufacturing a firm service by substituting across revocable quotes, but that statement awaits the first-position gate. The platform's allocation memory does not reward recent undercutting in the observed panel. Even within these boundaries, the inference router is a market designer: its admission, fallback, and steering rules determine the delivered product as much as the visible price surface does.

The resulting research agenda is concrete. Longer panels test whether gap-driven repricing and matched-menu excess survive new vintages. The randomized promotion gate estimates the deliverability frontier without order confounding. Provider- or router-side eligibility logs would separate rational quality protection from a tax on undercutting. Transparent decentralized markets supply an external laboratory in which accepted bids and termination reasons are public. Together, these designs can turn a market that currently displays menus into one whose commitments, allocation, and welfare can be audited.

References

Fernando Alvarez, Hervé Le Bihan, and Francesco Lippi. The real effects of monetary shocks in sticky price models: A sufficient statistic approach. *American Economic Review*, 106(10): 2817–2851, 2016. doi: 10.1257/aer.20140542.

Fernando E. Alvarez, Francesco Lippi, and Luigi Paciello. Optimal price setting with observation and menu costs. *Quarterly Journal of Economics*, 126(4):1909–1960, 2011. doi: 10.1093/qje/qjr043.

Iavor Bojinov, David Simchi-Levi, and Jinglong Zhao. Design and analysis of switchback experiments. *Management Science*, 69(7):3759–3777, 2023. doi: 10.1287/mnsc.2022.4583.

Zach Y. Brown and Alexander MacKay. Competition in pricing algorithms. *American Economic Journal: Microeconomics*, 15(2):109–156, 2023. doi: 10.1257/mic.20210158.

Brian Brubach, Nathaniel Grammel, Will Ma, and Aravind Srinivasan. Online matching frameworks under stochastic rewards, product ranking, and unknown patience. *Operations Research*, 73(2): 995–1010, 2025. doi: 10.1287/opre.2021.0371.

Guillermo A. Calvo. Staggered prices in a utility-maximizing framework. *Journal of Monetary Economics*, 12(3):383–398, 1983. doi: 10.1016/0304-3932(83)90060-0.

Lingjiao Chen, Matei Zaharia, and James Zou. Frugalgpt: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023. URL `https://arxiv.org/abs/2305.05176`.

David Holtz, Felipe Lobel, Ruben Lobel, Inessa Liskovich, and Sinan Aral. Reducing interference bias in online marketplace experiments using cluster randomization: Evidence from a pricing meta-experiment on Airbnb. *Management Science*, 71(1):390–406, 2025. doi: 10.1287/mnsc.2020.01157.

Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Kaustubh Upadhyay. Routerbench: A benchmark for multi-llm routing system. *arXiv preprint arXiv:2403.12031*, 2024. URL `https://arxiv.org/abs/2403.12031`.

Hugging Face. Inference providers: Provider selection and openai-compatible routing. Documentation, 2026. URL `https://huggingface.co/docs/inference-providers/en/index`. Accessed 2026-07-15; confirmatory policy suffixes frozen in the H89 preregistration.

Justin P. Johnson, Andrew Rhodes, and Matthijs Wildenbeest. Platform design when sellers use pricing algorithms. *Econometrica*, 91(5):1841–1879, 2023. doi: 10.3982/ECTA19978.

Christopher R. Knittel and Victor Stango. Price ceilings as focal points for tacit collusion: Evidence from credit cards. *American Economic Review*, 93(5):1703–1729, 2003. doi: 10.1257/000282803322655509.

Charles F. Manski. Identification of endogenous social effects: The reflection problem. *The Review of Economic Studies*, 60(3):531–542, 1993. doi: 10.2307/2298123.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. Routellm: Learning to route llms with preference data. *arXiv preprint arXiv:2406.18665*, 2024. URL `https://arxiv.org/abs/2406.18665`.

OpenRouter. Router metadata and generation records. Documentation, 2026. URL `https://openrouter.ai/docs/guides/features/router-metadata`. Accessed for the RouteScope protocol; source version must be pinned in a release manifest.

Zijie Wang. Who captures AI deflation? quality-adjusted prices, markups, and pass-through in AI services. Working paper, Warwick Business School, 2026. URL `https://zijie-cj-wang.github.io/home/`.

Xufeng Zhang, Shenghui Bao, and Han Li. Expert routing and self-preferencing on AI platforms. SSRN working paper 6404245, 2026.

A Variable construction and data-generating process

Variable	Construction
Completion quote	Output-token price in dollars per million tokens for the standard endpoint variant
Price event	First five-minute capture whose quote differs from the preceding same-endpoint capture
Cheapest rank	Dense rank of the workload-specific quote within model and capture time
Tie at minimum	At least two distinct providers have the exact minimum quote after common-unit conversion
Quote age	Time since the last observed event; first panel spell is left-censored
Recent cut	At least one negative completion-price event in the preceding seven calendar days
Default selection share	Selected-provider count divided by default probes in the provider-model-day cell
Rate-limit share	Router-published rate-limited count divided by successful plus rate-limited counts in the published window
Author-operated endpoint	Provider name matches the model author under the checked author-provider crosswalk
Direct parity	Direct API quote equals the contemporaneous router quote within machine tolerance

Table 8: Key variable definitions.

The capture scheduler targets five-minute cadence, but network and source failures create irregular intervals. All duration variables use actual UTC timestamps. Missing captures are never forward-filled for event timing. A quote that is unchanged across a gap contributes exposure but cannot locate an unobserved intermediate round trip; historical sources are therefore used as a lifecycle sensitivity, not to impute high-frequency paths.

The platform's five-minute and thirty-minute counters can overlap mechanically. Regressions use one horizon at a time. Capacity ceilings are displayed metadata, not contractual commitments, and missing ceilings are not coded as infinite. Derank transitions and rate-limit onsets are router enforcement events; they do not reveal which user's request caused the event.

B Proofs of the observational-equivalence results

Proof of proposition 1. For the nonreactive construction, let the public menu contain the finite union of price levels observed for the model. At each observed provider change, schedule that provider's private refresh clock between the two snapshots and let its state-dependent selector choose the observed terminal menu point. Providers that do not change retain their previous selection. The selector depends on the public menu state and provider state, not on a rival quote, yet reproduces the entire sampled panel. In particular, choosing a menu point already occupied by a rival creates both a cross-sectional match and a lagged exact landing without strategic response.

For the reactive construction, retain the same sampled terminal prices. For any chosen subset of the L exact-landing events, define the mover's response map at the realized history to select the matched rival quote; use the scheduled rule for all other events. This produces the same observations while classifying any integer number from zero through L as exact strategic landings. Dividing by N attains every grid point in $[0, L/N]$; mixtures attain the full interval. No event outside the L exact landings can enter the defined exact-landing estimand, proving sharpness. □

Proof of proposition 2. Without reactive replacement, $\Pr(Y_e = 1) = p_e$. A replacement forces $Y_e = 1$; otherwise the nonreactive draw remains. Hence $\Pr(Y_e = 1) = \rho + (1-\rho)p_e = p_e + \rho(1-p_e)$. Subtracting q_e gives equation (7). If $q_e \geq p_e$ and $p_e < 1$, solving the strict inequality $p_e - q_e + \rho(1 - p_e) > 0$ gives the stated threshold. Linearity of expectation gives the event-mean version when ρ is common and the event panel is fixed. □

Proof of proposition 3. Fix an observed transition and let S be the providers whose sampled quotes change. Choose $\epsilon > 0$ with $(|S| + 1)\epsilon < \Delta$. For a nonreactive path, order the providers in S arbitrarily and let provider i replace its old quote with its observed terminal quote at time $t_{k-1} + r_i\epsilon$. Declare these provider-specific update times and terminal quotes to be functions of the latent public state at the start of the interval. No rule uses a rival quote, and the sampled endpoints match the data.

For a reactive path, choose any provider $j \in S$ and any ordering of S with j first. Let j move at $t_{k-1} + \epsilon$. After the rth move, let the next provider observe that move and update at $t_{k-1} + (r + 1)\epsilon$; define its response map at this realized history to equal its sampled terminal quote. Every delay is strictly positive and less than Δ, and the terminal vector again matches the data. Repeating the construction with another j reverses the latent leader without changing either sampled endpoint. Providers outside S remain constant. Applying the construction interval by interval establishes the result for the full sampled panel. □

Proof of Corollary 4. An event with $U_e = 0$ cannot contribute to the named-rival estimand, and $R_e \in \{0, 1\}$, which gives equation (11). The lower endpoint is attained by the scheduled construction in proposition 3. For the upper endpoint, use the reactive construction for every $U_e = 1$ event, assigning its unique strictly prior rival as the trigger; unrestricted response maps preserve all sampled terminal quotes. Thus neither endpoint can be tightened from the snapshot panel alone. □

The proposition is intentionally nonparametric. It does not say scheduled and reactive models remain equivalent after imposing a known common-state process, a parametric reaction map, a lower bound on response latency, or continuous update timestamps. It says those restrictions are additional identifying data or assumptions; they are not supplied by a longer panel at the same observation frequency.

C Independent grid-null algorithm

For each observed model-day (m, d) with $N_{md} \geq 2$:

1. Compute the observed median log price $\widetilde{p}_{md}$ and provider deviations $u_{imd} = \log p_{imd} - \widetilde{p}_{md}$.
2. Pool deviations over model-days after the same endpoint and variant filters used by the observed statistic.
3. Draw N_{md} deviations independently with replacement, add $\widetilde{p}_{md}$, exponentiate, and snap to the declared grid.
4. Record whether the two smallest simulated prices are equal. Aggregate over model-days and repeat the full simulation.

The primary grid is one cent per million tokens; the sensitivity is ten cents. The simulation preserves the distribution of market thickness and model-day price levels but removes cross-provider dependence. Because the deviation pool is empirical, it does not assume a parametric cost distribution. Its main limitation is that pooled deviations combine provider heterogeneity and may retain atoms from the observed coordination process.

D Randomized-crossover protocol

Each eligible block chooses a hot model with at least three publicly quoted providers. It freezes the contemporaneous provider set and computes cheapest, second-cheapest, and random pinned candidates. A deterministic pseudorandom seed permutes the four policies and the model order. The assignment row is written before requests begin. Requests use a fixed minimal prompt and one output token.

The primary outcomes are HTTP-level success and a valid generation identifier. Secondary outcomes are latency, observed spend, selected provider, and fallback attempts when exposed. Missing cost is not recoded as zero. A block is valid only if assignment replay passes, the quote snapshot predates the request, and each declared policy is attempted exactly once. The confirmatory prefix stops at 500 valid attempts per pinned arm; later rows are a separate continuation sample.

The three primary hypotheses compare default success with each pinned arm. The family uses Holm adjustment. Randomization inference permutes policy labels according to the original assignment mechanism within blocks. The first-position estimator is primary for arbitrary carryover. Pooled position-adjusted contrasts and model-policy heterogeneity are secondary. A flat cheapest/second/random gradient is reported as evidence against rank-specific stale-quote refusal, not as equivalence unless the registered equivalence margin is met.

E Steering-audit robustness

The unrestricted audit includes every provider-model-day that is currently cheapest on the public surface and has at least one default probe. Model-day fixed effects remove common demand and quote-level shocks; price-rank controls compare providers at the same current prominence. Standard errors cluster at provider and model-day in the promotion specification.

Three robustness layers are preregistered: (i) require at least one successful pinned request for the provider-model pair; (ii) use only regions and prompt lengths for which all compared providers pass eligibility; and (iii) instrument recent-cut visibility with randomized quote-surface exposure if the platform permits a safe visibility experiment. Current data support the direction in layer (i) but not its sample size. No visibility instrument has fired.

F Transparent-compute comparator

A separately preregistered Akash study links public multi-provider bid sets, the accepted lease, and exact on-chain termination reasons. The launch backfill replayed 2,929 retained closed leases with a 100% exact-ID match. Its pre-preregistration support set contains 36 linked multi-provider choices: 20 accepted prices above the lowest bid and 16 at the lowest. All 36 have the fixed 300-block follow-up, but only 20 close in that window; none records a provider, escrow, or non-owner close. This is zero-event support, not evidence of no effect.

The prospective cohort uses a second-snapshot inception rule, masks confirmatory outcomes, and has a fixed release of 2026-08-15. The comparator is excluded from the paper's promoted facts because its market object is a compute lease rather than a token-priced inference request. Its role is methodological: it shows what becomes identifiable when bids, acceptance, and termination are public.

G Claim ledger and promotion gates

Claim	Current evidence	Missing identification	Promotion gate
Administered menus	Frequency, size, grid, clock, day-split gap AUC	Long-run frailty and strategic response	30 panel days, both vintages
Exact price atoms	Tie atom survives independent grid null; author null fails; lagged-landing test is nonrejecting and underpowered	Anchor identity and strategic response	30-date endpoint-label and matched-menu replication
Manufactured firmness	Secondary randomized-order block level, flat rank gradient	Carryover-robust first-position estimate and transport	500 first-position assignments per arm
Negative steering memory	Rank-conditioned audit, same-sign eligibility subset	Complete eligibility or visibility instrument	Eligibility-complete panel
Brown–MacKay mechanism	Cadence premium; reaction rules predictively indistinguishable	Slow-initiator/fast-responder exposure	Earliest 30-day vintage with fixed paired test
Retry externality	Forward/backward asymmetry, small IV estimate	Cross-provider retry and user value	Incident instrument plus attempt logs
Literal front-running	Proposition 3; named-rival response share in $[0, 68.6\%]$	Cross-user order and quote-update sequence	Negotiated router/provider event logs
Global welfare loss	None	Values, quality, costs, transfers, capacity response	Structural measurement contract

Table 9: The ledger distinguishes a supported market fact from the stronger mechanism or welfare claim it may motivate.

H Reproducibility

All tables come from versioned Parquet and JSON artifacts carrying source and retrieval timestamps and run identifiers. Probe assignments record study and block identifiers, seed, intended policy, position, and quote-snapshot key; prompts and secrets are excluded. The repository includes analyzers, preregistrations, claim boundaries, and machine-readable summaries. The four-panel figure reads only frozen `paper_estimates.json`, not the pre-randomization legacy H80 summary. The manuscript release resolves the Hugging Face dataset head before its first query and pins every subsequent query to that immutable revision. The present nine-date release uses commit `600bb41fd15189c70f8f78fce8cf0a519fb8dd61`; an independent pinned replay reproduced all 16 published artifact hashes. This distinction matters because a date window alone does not prevent a late backfill from changing an earlier date. The outcome-free `manuscript_promotion_gate_summary.json` freezes the nine-day and earliest 30-day vintages and masks H80 outcomes until 500 first-position assignments per arm. The companion `manuscript_vintages` program runs PM1, PM5, and BM1–BM4 under both calendar cutoffs, canonicalizes every table, records SHA-256 hashes, and compares only the metrics fixed before the 30-day data exist. Each gate release updates estimates, source manifest, and PDF in one commit.

Privacy and platform load. Capture records exclude prompts and completions. The owned experiments use a fixed minimal prompt, cap output at one token, serialize potentially interfering studies, and keep spend bounded by construction. Public quote requests and router-published counters are rate limited; failed captures create missing intervals rather than immediate retry storms.

Release tiers. The public replication tier contains code, public-source manifests, quote-derived tables, randomized assignment fields, and aggregated confirmatory statistics. Account-specific generation references and raw route attempts remain in the restricted evidence store. The release exposes the seeds and fields needed to replay assignment and hashes every restricted-input aggregate, without exposing request payloads or credentials.